

Movie Booking System (BookMyShow)

Patterns: State Pattern, Strategy Pattern (Pricing), Singleton Pattern, Lock Manager | Domain: Resource Booking / Concurrency Lock

Core Requirements & Features	Core Domain Classes & Interfaces
• Concert/Movie Booking: Browse movies, shows, halls, & seat availability. • Concurrent Seat Locks: Temporary 10-minute seat lock during checkout. • Payment & Ticket: Process payment, confirm booking, and generate QR ticket.	• MovieBookingService: Facade managing shows, seat maps, & bookings. • SeatLockManager: Handles temporary seat locks with TTL expiration. • Show & Seat: Showtime metadata, seat status (AVAILABLE, LOCKED, BOOKED). • PricingStrategy: Dynamic pricing based on seat tier, time, & demand.

Critical Execution Workflow & Method Trace

1. User selects seats [A1, A2] for Show 101.
 2. MovieBookingService invokes SeatLockManager.lockSeats(showId, seats, userId).
 3. SeatLockManager acquires ReentrantLock for Show 101.
 4. Validates all requested seats are currently in AVAILABLE status.
 5. Updates seat status to LOCKED, records timestamp, and schedules 10-minute expiration timer (ScheduledExecutorService / Redis TTL).
 6. User completes payment before lock expiry: SeatLockManager promotes status to BOOKED, creates Ticket, returns Confirmation.
 7. If payment fails or timer expires: SeatLockManager unlocks seats back to AVAILABLE.

Concurrency & Thread Safety Mechanics	Design Trade-offs & Interview FAQs
• Double-Booking Defense: Explicit per-Show ReentrantLock or DB row SELECT FOR UPDATE. • Lock Expiration: Atomic expiration handling prevents deadlocks when users abandon carts. • Idempotence: Payment webhook verifies booking ID state to prevent double ticket generation.	• Pessimistic vs Optimistic: Pessimistic Seat Locking prevents user frustration during high-demand movie launches. • Distributed Locks: Scale from in-memory ReentrantLock to Redis Redlock for multi-instance deployments. • Decoupling: PricingStrategy decoupled from booking flow to support dynamic surging.